

CS2206 - Computer Architecture Project

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Two-Pass Assembler and Emulator for the SIMPLEX Instruction Set

1. Compiling assembler and emulator

```
PS C:\Users\saich\OneDrive\Desktop\AI> gcc asm.c -o asm
PS C:\Users\saich\OneDrive\Desktop\AI> gcc emu.c -o emu
```

2. test01.asm and test01.o

```
PS C:\Users\saich\OneDrive\Desktop\AI> ./asm test01.asm
WARNING line 0: label 'label' defined but never used
Object : test01.o (8 words)
Listing : test01.lst
Result : 0 error(s), 1 warning(s)
PS C:\Users\saich\OneDrive\Desktop\AI> ./emu test01.o
Loaded 8 instructions
ERROR: instruction limit exceeded (infinite loop?)
```

3. test01.log

```
test01.log
1  WARNING line 0: label 'label' defined but never used
2
3  --- 0 error(s), 1 warning(s) ---
4
```

4. test01.lst

```
test01.lst
1  00000000 label:
2  00000000 00000000 ldc 0
3  00000001 FFFFFFFB00 ldc -5
4  00000002 00000500 ldc 5
5  00000003 loop:
6  00000003 FFFFFFFF11 br loop
7  00000004 00000011 br next
8  00000005 next:
9  00000005 00000300 ldc loop
10 00000006 00000700 ldc var1
11 00000007 var1:
12 00000007 00000000 data 0
13 ; WARNING: label 'label' defined but never used
14
```

5. test02.asm and test02.o

```
PS C:\Users\saich\OneDrive\Desktop\AI> ./asm test02.asm
ERROR line 8: duplicate label 'label'
ERROR line 10: invalid operand '08ge'
ERROR line 11: 'ldc' requires an operand
ERROR line 12: 'add' takes no operand (found '5')
ERROR line 13: unexpected text after operand: ', 6; extra on end of line '
ERROR line 14: invalid label name '0def'
ERROR line 15: unknown mnemonic 'fibble'
ERROR line 16: unknown mnemonic '0def'
ERROR line 9: undefined label 'nonesuch'
ERROR line 10: invalid operand '08ge'
ERROR line 11: 'ldc' requires an operand
ERROR line 12: 'add' takes no operand (found '5')
ERROR line 13: unexpected text after operand: ', 6; extra on end of line '
ERROR line 14: invalid label name '0def'
ERROR line 15: unknown mnemonic 'fibble'
ERROR line 16: unknown mnemonic '0def'
WARNING line 0: label 'label' defined but never used
Assembly failed - no output files produced.
Result : 16 error(s), 1 warning(s)
PS C:\Users\saich\OneDrive\Desktop\AI> ./emu test02.o
Cannot open 'test02.o'
```

6. test02.log

```
test02.log
1 ERROR line 8: duplicate label 'label'
2 ERROR line 10: invalid operand '08ge'
3 ERROR line 11: 'ldc' requires an operand
4 ERROR line 12: 'add' takes no operand (found '5')
5 ERROR line 13: unexpected text after operand: ', 6; extra on end of line '
6 ERROR line 14: invalid label name '0def'
7 ERROR line 15: unknown mnemonic 'fibble'
8 ERROR line 16: unknown mnemonic '0def'
9 ERROR line 9: undefined label 'nonesuch'
10 ERROR line 10: invalid operand '08ge'
11 ERROR line 11: 'ldc' requires an operand
12 ERROR line 12: 'add' takes no operand (found '5')
13 ERROR line 13: unexpected text after operand: ', 6; extra on end of line '
14 ERROR line 14: invalid label name '0def'
15 ERROR line 15: unknown mnemonic 'fibble'
16 ERROR line 16: unknown mnemonic '0def'
17 WARNING line 0: label 'label' defined but never used
18
19 --- 16 error(s), 1 warning(s) ---
20
```

7. test03.asm and test03.o

```
PS C:\Users\saich\OneDrive\Desktop\AI> ./asm test03.asm
Object : test03.o (3 words)
Listing : test03.lst
Result : 0 error(s), 0 warning(s)
PS C:\Users\saich\OneDrive\Desktop\AI> ./emu test03.o
Loaded 3 instructions

HALT reached
Final Registers:
A = 141
B = 0
PC = 3
SP = 0

--- Memory Dump (non-zero locations) ---
memory[0] = 19200 (0x00004B00)
memory[1] = 16897 (0x00004201)
memory[2] = 18 (0x00000012)
```

8. test03.log

```
test03.log
1
2   --- 0 error(s), 0 warning(s) ---
3   |
```

9. test03.lst

```
test03.lst
1   00000000 val:
2   00000000 00004B00   ldc val
3   00000001 00004201   adc val2
4   00000002 00000012   HALT
5   00000003 val2:
6
```

10. triangle.asm and triangle.o

```
PS C:\Users\saich\OneDrive\Desktop\AI> ./asm triangle.asm
Object : triangle.o (76 words)
Listing : triangle.lst
Result : 0 error(s), 0 warning(s)
```

```
PS C:\Users\saich\OneDrive\Desktop\AI> ./emu triangle.o
Loaded 76 instructions
```

```
HALT reached
```

```
Final Registers:
```

```
A = 0
```

```
B = 0
```

```
PC = 10
```

```
SP = 4096
```

```
--- Memory Dump (non-zero locations) ---
```

```
memory[0] = 1048576 (0x00100000)
memory[1] = 11 (0x0000000B)
memory[2] = -246 (0xFFFFF0A)
memory[3] = 19200 (0x00004B00)
memory[4] = 3 (0x00000003)
memory[5] = 18944 (0x00004A00)
memory[6] = 4 (0x00000004)
memory[7] = 525 (0x0000020D)
memory[8] = 266 (0x0000010A)
memory[9] = 18 (0x00000012)
memory[10] = -758 (0xFFFFD0A)
memory[11] = 259 (0x00000103)
memory[12] = 515 (0x00000203)
memory[14] = 3 (0x00000003)
memory[15] = -246 (0xFFFFF0A)
memory[16] = 770 (0x00000302)
memory[17] = 3 (0x00000003)
memory[18] = 258 (0x00000102)
memory[19] = 4109 (0x0000100D)
memory[20] = 266 (0x0000010A)
memory[21] = 770 (0x00000302)
memory[22] = 5 (0x00000005)
memory[23] = 770 (0x00000302)
memory[24] = 257 (0x00000101)
memory[25] = 771 (0x00000303)
memory[26] = 2 (0x00000002)
memory[27] = 257 (0x00000101)
memory[28] = 3 (0x00000003)
memory[29] = 2 (0x00000002)
memory[30] = 514 (0x00000202)
memory[31] = 7 (0x00000007)
memory[32] = -4592 (0xFFFFEE10)
memory[33] = 258 (0x00000102)
memory[34] = 778 (0x0000030A)
memory[35] = 14 (0x0000000E)
```

```
memory[36] = -758 (0xFFFFD0A)
memory[37] = 259 (0x00000103)
memory[38] = 515 (0x00000203)
memory[39] = 256 (0x00000100)
memory[40] = 8 (0x00000008)
memory[41] = 770 (0x00000302)
memory[42] = 7 (0x00000007)
memory[43] = 1040 (0x00000410)
memory[44] = 770 (0x00000302)
memory[45] = 514 (0x00000202)
memory[46] = 7 (0x00000007)
memory[47] = 515 (0x00000203)
memory[48] = 514 (0x00000202)
memory[49] = 5135 (0x0000140F)
memory[50] = 770 (0x00000302)
memory[51] = -255 (0xFFFFF01)
memory[52] = 3 (0x00000003)
memory[53] = -246 (0xFFFFF0A)
memory[54] = 258 (0x00000102)
memory[55] = 3 (0x00000003)
memory[56] = 770 (0x00000302)
memory[57] = -255 (0xFFFFF01)
memory[58] = -5875 (0xFFFFE90D)
memory[59] = 258 (0x00000102)
memory[60] = 3 (0x00000003)
memory[61] = 259 (0x00000103)
memory[62] = 770 (0x00000302)
memory[63] = -7155 (0xFFFFE40D)
memory[64] = 266 (0x0000010A)
memory[65] = 2 (0x00000002)
memory[66] = 6 (0x00000006)
memory[67] = 258 (0x00000102)
memory[68] = 778 (0x0000030A)
memory[69] = 14 (0x0000000E)
memory[70] = 256 (0x00000100)
memory[71] = 258 (0x00000102)
memory[72] = 778 (0x0000030A)
memory[73] = 14 (0x0000000E)
memory[74] = 10 (0x0000000A)
memory[75] = 1 (0x00000001)
memory[76] = 10 (0x0000000A)
memory[77] = 45 (0x0000002D)
memory[78] = 120 (0x00000078)
memory[79] = 210 (0x000000D2)
memory[80] = 252 (0x000000FC)
memory[81] = 210 (0x000000D2)
```

```
memory[82] = 120 (0x00000078)
memory[83] = 45 (0x0000002D)
memory[84] = 10 (0x0000000A)
memory[4053] = 64 (0x00000040)
memory[4055] = 1 (0x00000001)
memory[4056] = 1 (0x00000001)
memory[4057] = 64 (0x00000040)
memory[4058] = 1 (0x00000001)
memory[4059] = 2 (0x00000002)
memory[4060] = 1 (0x00000001)
memory[4061] = 64 (0x00000040)
memory[4062] = 1 (0x00000001)
memory[4063] = 3 (0x00000003)
memory[4064] = 1 (0x00000001)
memory[4065] = 64 (0x00000040)
memory[4066] = 1 (0x00000001)
memory[4067] = 4 (0x00000004)
memory[4068] = 1 (0x00000001)
memory[4069] = 64 (0x00000040)
memory[4070] = 1 (0x00000001)
memory[4071] = 5 (0x00000005)
memory[4072] = 1 (0x00000001)
memory[4073] = 64 (0x00000040)
memory[4074] = 1 (0x00000001)
memory[4075] = 6 (0x00000006)
memory[4076] = 1 (0x00000001)
memory[4077] = 64 (0x00000040)
memory[4078] = 1 (0x00000001)
memory[4079] = 7 (0x00000007)
memory[4080] = 1 (0x00000001)
memory[4081] = 64 (0x00000040)
memory[4082] = 1 (0x00000001)
memory[4083] = 8 (0x00000008)
memory[4084] = 1 (0x00000001)
memory[4085] = 64 (0x00000040)
memory[4086] = 1 (0x00000001)
memory[4087] = 9 (0x00000009)
memory[4088] = 1 (0x00000001)
memory[4089] = 20 (0x00000014)
memory[4090] = 1 (0x00000001)
memory[4091] = 10 (0x0000000A)
memory[4092] = 10 (0x0000000A)
memory[4093] = 8 (0x00000008)
memory[4094] = 10 (0x0000000A)
memory[4095] = 85 (0x00000055)
```

11. triangle.log

```
≡ triangle.log
1
2   --- 0 error(s), 0 warning(s) ---
3
```

12. triangle.lst

```
≡ triangle.lst
1  00000000 00100000 ldc 4096
2  00000001 0000000B a2sp
3  00000002 FFFFFFF0A adj -1
4  00000003 00004B00 ldc result
5  00000004 00000003 stl 0
6  00000005 00004A00 ldc count
7  00000006 00000004 ldnl 0
8  00000007 0000020D call main
9  00000008 0000010A adj 1
10 00000009 00000012 HALT
11 0000000A main:
12 0000000A FFFFFD0A adj -3
13 0000000B 00000103 stl 1
14 0000000C 00000203 stl 2
15 0000000D 00000000 ldc 0
16 0000000E 00000003 stl 0
17 0000000F loop:
18 0000000F FFFFFFF0A adj -1
19 00000010 00000302 ldl 3
20 00000011 00000003 stl 0
21 00000012 00000102 ldl 1
22 00000013 0000100D call triangle
23 00000014 0000010A adj 1
24 00000015 00000302 ldl 3
25 00000016 00000005 stnl 0
26 00000017 00000302 ldl 3
27 00000018 00000101 adc 1
28 00000019 00000303 stl 3
29 0000001A 00000002 ldl 0
30 0000001B 00000101 adc 1
31 0000001C 00000003 stl 0
32 0000001D 00000002 ldl 0
33 0000001E 00000202 ldl 2
34 0000001F 00000007 sub
35 00000020 FFFFEE10 brlz loop
36 00000021 00000102 ldl 1
37 00000022 0000030A adj 3
38 00000023 0000000E return
39 00000024 triangle:
40 00000024 FFFFFD0A adj -3
41 00000025 00000103 stl 1
42 00000026 00000203 stl 2
```

triangle.lst

```
43  00000027 00000100 ldc 1
44  00000028 00000008 shl
45  00000029 00000302 ldl 3
46  0000002A 00000007 sub
47  0000002B 00000410 brlz skip
48  0000002C 00000302 ldl 3
49  0000002D 00000202 ldl 2
50  0000002E 00000007 sub
51  0000002F 00000203 stl 2
52  00000030 skip:
53  00000030 00000202 ldl 2
54  00000031 0000140F brz one
55  00000032 00000302 ldl 3
56  00000033 FFFFFFF01 adc -1
57  00000034 00000003 stl 0
58  00000035 FFFFFFF0A adj -1
59  00000036 00000102 ldl 1
60  00000037 00000003 stl 0
61  00000038 00000302 ldl 3
62  00000039 FFFFFFF01 adc -1
63  0000003A FFFFE90D call triangle
64  0000003B 00000102 ldl 1
65  0000003C 00000003 stl 0
66  0000003D 00000103 stl 1
67  0000003E 00000302 ldl 3
68  0000003F FFFFE40D call triangle
69  00000040 0000010A adj 1
70  00000041 00000002 ldl 0
71  00000042 00000006 add
72  00000043 00000102 ldl 1
73  00000044 0000030A adj 3
74  00000045 0000000E return
75  00000046 one:
76  00000046 00000100 ldc 1
77  00000047 00000102 ldl 1
78  00000048 0000030A adj 3
79  00000049 0000000E return
80  0000004A count:
81  0000004A 0000000A data 10
82  0000004B result:
83  0000004B 00000000 data 0
```


13. bubble_sort.asm and bubble_sort.o

```
PS C:\Users\saich\OneDrive\Desktop\AI> ./asm bubble_sort.asm
Object : bubble_sort.o (77 words)
Listing : bubble_sort.lst
Result : 0 error(s), 0 warning(s)
PS C:\Users\saich\OneDrive\Desktop\AI> ./emu bubble_sort.o
Loaded 77 instructions

HALT reached
Final Registers:
A = 0
B = 0
PC = 6
SP = 4096

--- Memory Dump (non-zero locations) ---
memory[0] = 1048576 (0x00100000)
memory[1] = 11 (0x0000000B)
memory[2] = -246 (0xFFFFF0A)
memory[3] = 525 (0x0000020D)
memory[4] = 266 (0x0000010A)
memory[5] = 18 (0x00000012)
memory[6] = -1782 (0xFFFF90A)
memory[7] = 259 (0x00000103)
memory[9] = 515 (0x00000203)
memory[10] = 17408 (0x00004400)
memory[11] = 4 (0x00000004)
memory[12] = -255 (0xFFFFF01)
memory[13] = 3 (0x00000003)
memory[14] = 514 (0x00000202)
memory[15] = 2 (0x00000002)
memory[16] = 7 (0x00000007)
memory[17] = 272 (0x00000110)
memory[18] = 11793 (0x00002E11)
memory[20] = 771 (0x00000303)
memory[21] = 17408 (0x00004400)
memory[22] = 4 (0x00000004)
memory[23] = -255 (0xFFFFF01)
memory[24] = 3 (0x00000003)
memory[25] = 514 (0x00000202)
memory[26] = 2 (0x00000002)
memory[27] = 7 (0x00000007)
memory[28] = 770 (0x00000302)
memory[29] = 6 (0x00000006)
memory[30] = 272 (0x00000110)
memory[31] = 7441 (0x00001D11)
memory[32] = 17664 (0x00004500)
```

```
memory[33] = 770 (0x00000302)
memory[34] = 6 (0x00000006)
memory[35] = 1283 (0x00000503)
memory[36] = 17664 (0x00004500)
memory[37] = 770 (0x00000302)
memory[38] = 257 (0x00000101)
memory[39] = 6 (0x00000006)
memory[40] = 1539 (0x00000603)
memory[41] = 1282 (0x00000502)
memory[42] = 4 (0x00000004)
memory[43] = 1027 (0x00000403)
memory[44] = 1538 (0x00000602)
memory[45] = 4 (0x00000004)
memory[46] = 1026 (0x00000402)
memory[47] = 7 (0x00000007)
memory[48] = 272 (0x00000110)
memory[49] = 1809 (0x00000711)
memory[50] = 1538 (0x00000602)
memory[51] = 4 (0x00000004)
memory[52] = 1282 (0x00000502)
memory[53] = 5 (0x00000005)
memory[54] = 1026 (0x00000402)
memory[55] = 1538 (0x00000602)
memory[56] = 5 (0x00000005)
memory[57] = 770 (0x00000302)
memory[58] = 257 (0x00000101)
memory[59] = 771 (0x00000303)
memory[60] = -10223 (0xFFFFD811)
memory[61] = 514 (0x00000202)
memory[62] = 257 (0x00000101)
memory[63] = 515 (0x00000203)
memory[64] = -14063 (0xFFFFC911)
memory[65] = 258 (0x00000102)
memory[66] = 1802 (0x0000070A)
memory[67] = 14 (0x0000000E)
memory[68] = 8 (0x00000008)
memory[69] = 3 (0x00000003)
memory[70] = 12 (0x0000000C)
memory[71] = 19 (0x00000013)
memory[72] = 25 (0x00000019)
memory[73] = 45 (0x0000002D)
memory[74] = 64 (0x00000040)
memory[75] = 78 (0x0000004E)
memory[76] = 87 (0x00000057)
memory[4088] = 7 (0x00000007)
memory[4089] = 4 (0x00000004)
```

```
memory[4090] = 7 (0x00000007)
memory[4091] = 1 (0x00000001)
memory[4092] = 3 (0x00000003)
memory[4093] = 69 (0x00000045)
memory[4094] = 70 (0x00000046)
```

14. bubble_sort.log

```
≡ bubble_sort.log
1
2 --- 0 error(s), 0 warning(s) ---
3
```

15. bubble_sort.lst

```
≡ bubble_sort.lst
1 00000000 00100000 ldc 4096
2 00000001 0000000B a2sp
3 00000002 FFFFFFF0A adj -1
4 00000003 0000020D call sort
5 00000004 0000010A adj 1
6 00000005 00000012 HALT
7 00000006 sort:
8 00000006 FFFFF90A adj -7
9 00000007 00000103 stl 1
10 00000008 00000000 ldc 0
11 00000009 00000203 stl 2
12 0000000A outer:
13 0000000A 00004400 ldc N
14 0000000B 00000004 ldnl 0
15 0000000C FFFFFFF01 adc -1
16 0000000D 00000003 stl 0
17 0000000E 00000202 ldl 2
18 0000000F 00000002 ldl 0
19 00000010 00000007 sub
20 00000011 00000110 brlz outer_body
21 00000012 00002E11 br sort_done
22 00000013 outer_body:
23 00000013 00000000 ldc 0
24 00000014 00000303 stl 3
25 00000015 inner:
26 00000015 00004400 ldc N
27 00000016 00000004 ldnl 0
28 00000017 FFFFFFF01 adc -1
29 00000018 00000003 stl 0
30 00000019 00000202 ldl 2
31 0000001A 00000002 ldl 0
32 0000001B 00000007 sub
33 0000001C 00000302 ldl 3
34 0000001D 00000006 add
35 0000001E 00000110 brlz inner_body
36 0000001F 00001D11 br next_outer
37 00000020 inner_body:
38 00000020 00004500 ldc array
39 00000021 00000302 ldl 3
40 00000022 00000006 add
41 00000023 00000503 stl 5
42 00000024 00004500 ldc array
```

```

43  00000025 00000302  ldl 3
44  00000026 00000101  adc 1
45  00000027 00000006  add
46  00000028 00000603  stl 6
47  00000029 00000502  ldl 5
48  0000002A 00000004  ldnl 0
49  0000002B 00000403  stl 4
50  0000002C 00000602  ldl 6
51  0000002D 00000004  ldnl 0
52  0000002E 00000402  ldl 4
53  0000002F 00000007  sub
54  00000030 00000110  brlz do_swap
55  00000031 00000711  br no_swap
56  00000032 do_swap:
57  00000032 00000602  ldl 6
58  00000033 00000004  ldnl 0
59  00000034 00000502  ldl 5
60  00000035 00000005  stnl 0
61  00000036 00000402  ldl 4
62  00000037 00000602  ldl 6
63  00000038 00000005  stnl 0
64  00000039 no_swap:
65  00000039 00000302  ldl 3
66  0000003A 00000101  adc 1
67  0000003B 00000303  stl 3
68  0000003C FFFFD811  br inner
69  0000003D next_outer:
70  0000003D 00000202  ldl 2
71  0000003E 00000101  adc 1
72  0000003F 00000203  stl 2
73  00000040 FFFFC911  br outer
74  00000041 sort_done:
75  00000041 00000102  ldl 1
76  00000042 0000070A  adj 7
77  00000043 0000000E  return
78  00000044 N:
79  00000044 00000008  data 8
80  00000045 array:
81  00000045 00000040  data 64
82  00000046 00000019  data 25
83  00000047 0000000C  data 12
84  00000048 00000057  data 87

```

```

85  00000049 00000003  data 3
86  0000004A 0000002D  data 45
87  0000004B 0000004E  data 78
88  0000004C 00000013  data 19

```

16. factorial.asm and factorial.o

```
PS C:\Users\saich\OneDrive\Desktop\AI> ./asm factorial.asm
Object : factorial.o (51 words)
Listing : factorial.lst
Result : 0 error(s), 0 warning(s)
PS C:\Users\saich\OneDrive\Desktop\AI> ./emu factorial.o
Loaded 51 instructions

HALT reached
Final Registers:
A = 120
B = 50
PC = 8
SP = 4096

--- Memory Dump (non-zero locations) ---
memory[0] = 1048576 (0x00100000)
memory[1] = 11 (0x0000000B)
memory[2] = -246 (0xFFFFF0A)
memory[3] = 1037 (0x0000040D)
memory[4] = 266 (0x0000010A)
memory[5] = 12800 (0x00003200)
memory[6] = 4 (0x00000004)
memory[7] = 18 (0x00000012)
memory[8] = -1270 (0xFFFFB0A)
memory[9] = 259 (0x00000103)
memory[10] = 512 (0x00000200)
memory[11] = 515 (0x00000203)
memory[12] = 256 (0x00000100)
memory[13] = 771 (0x00000303)
memory[14] = 12544 (0x00003100)
memory[15] = 4 (0x00000004)
memory[16] = 3 (0x00000003)
memory[17] = 514 (0x00000202)
memory[18] = 2 (0x00000002)
memory[19] = 7 (0x00000007)
memory[20] = 528 (0x00000210)
memory[21] = 271 (0x0000010F)
memory[22] = 5137 (0x00001411)
memory[24] = 1283 (0x00000503)
memory[25] = 514 (0x00000202)
memory[26] = 1027 (0x00000403)
memory[27] = 1026 (0x00000402)
memory[28] = 2063 (0x0000080F)
memory[29] = 1282 (0x00000502)
memory[30] = 770 (0x00000302)
memory[31] = 6 (0x00000006)
```

```
memory[32] = 1283 (0x00000503)
memory[33] = 1026 (0x00000402)
memory[34] = -255 (0xFFFFFFFF01)
memory[35] = 1027 (0x00000403)
memory[36] = -2543 (0xFFFFF611)
memory[37] = 1282 (0x00000502)
memory[38] = 771 (0x00000303)
memory[39] = 514 (0x00000202)
memory[40] = 257 (0x00000101)
memory[41] = 515 (0x00000203)
memory[42] = -7407 (0xFFFFE311)
memory[43] = 770 (0x00000302)
memory[44] = 12800 (0x00003200)
memory[45] = 5 (0x00000005)
memory[46] = 258 (0x00000102)
memory[47] = 1290 (0x0000050A)
memory[48] = 14 (0x0000000E)
memory[49] = 5 (0x00000005)
memory[50] = 120 (0x00000078)
memory[4090] = 5 (0x00000005)
memory[4091] = 4 (0x00000004)
memory[4092] = 6 (0x00000006)
memory[4093] = 120 (0x00000078)
memory[4095] = 120 (0x00000078)
```

17. factorial.log

```
factorial.log
1
2 --- 0 error(s), 0 warning(s) ---
3
```

18. factorial.lst

```
≡ factorial.lst
1  00000000 00100000 ldc 4096
2  00000001 0000000B a2sp
3  00000002 FFFFFFF0A adj -1
4  00000003 0000040D call main
5  00000004 0000010A adj 1
6  00000005 00003200 ldc result
7  00000006 00000004 ldnl 0
8  00000007 00000012 HALT
9  00000008 main:
10 00000008 FFFFFB0A adj -5
11 00000009 00000103 stl 1
12 0000000A 00000200 ldc 2
13 0000000B 00000203 stl 2
14 0000000C 00000100 ldc 1
15 0000000D 00000303 stl 3
16 0000000E outer:
17 0000000E 00003100 ldc N
18 0000000F 00000004 ldnl 0
19 00000010 00000003 stl 0
20 00000011 00000202 ldl 2
21 00000012 00000002 ldl 0
22 00000013 00000007 sub
23 00000014 00000210 brlz do_mul
24 00000015 0000010F brz do_mul
25 00000016 00001411 br outer_done
26 00000017 do_mul:
27 00000017 00000000 ldc 0
28 00000018 00000503 stl 5
29 00000019 00000202 ldl 2
30 0000001A 00000403 stl 4
31 0000001B inner:
32 0000001B 00000402 ldl 4
33 0000001C 0000080F brz mul_done
34 0000001D 00000502 ldl 5
35 0000001E 00000302 ldl 3
36 0000001F 00000006 add
37 00000020 00000503 stl 5
38 00000021 00000402 ldl 4
39 00000022 FFFFFFF01 adc -1
40 00000023 00000403 stl 4
41 00000024 FFFFF611 br inner
42 00000025 mul_done:
```

```

43  00000025 00000502  ldl 5
44  00000026 00000303  stl 3
45  00000027 00000202  ldl 2
46  00000028 00000101  adc 1
47  00000029 00000203  stl 2
48  0000002A FFFFE311  br outer
49  0000002B outer_done:
50  0000002B 00000302  ldl 3
51  0000002C 00003200  ldc result
52  0000002D 00000005  stnl 0
53  0000002E 00000102  ldl 1
54  0000002F 0000050A  adj 5
55  00000030 0000000E  return
56  00000031 N:
57  00000031 00000005  data 5
58  00000032 result:
59  00000032 00000000  data 0
60

```

19. fibonacci.asm and fibonacci.o

```

PS C:\Users\saich\OneDrive\Desktop\AI> ./asm fibonacci.asm
Object : fibonacci.o (37 words)
Listing : fibonacci.lst
Result : 0 error(s), 0 warning(s)
PS C:\Users\saich\OneDrive\Desktop\AI> ./emu fibonacci.o
Loaded 37 instructions

HALT reached
Final Registers:
A = 55
B = 36
PC = 8
SP = 4096

```



```
--- Memory Dump (non-zero locations) ---
memory[0] = 1048576 (0x00100000)
memory[1] = 11 (0x0000000B)
memory[2] = -246 (0xFFFFF0A)
memory[3] = 1037 (0x0000040D)
memory[4] = 266 (0x0000010A)
memory[5] = 9216 (0x00002400)
memory[6] = 4 (0x00000004)
memory[7] = 18 (0x00000012)
memory[8] = -1014 (0xFFFFC0A)
memory[9] = 259 (0x00000103)
memory[10] = 8960 (0x00002300)
memory[11] = 4 (0x00000004)
memory[12] = -255 (0xFFFFF01)
memory[13] = 515 (0x00000203)
memory[15] = 771 (0x00000303)
memory[16] = 256 (0x00000100)
memory[17] = 1027 (0x00000403)
memory[18] = 514 (0x00000202)
memory[19] = 2319 (0x0000090F)
memory[20] = 1026 (0x00000402)
memory[21] = 770 (0x00000302)
memory[22] = 6 (0x00000006)
memory[23] = 1027 (0x00000403)
memory[24] = 771 (0x00000303)
memory[25] = 514 (0x00000202)
memory[26] = -255 (0xFFFFF01)
memory[27] = 515 (0x00000203)
memory[28] = -2799 (0xFFFFF511)
memory[29] = 1026 (0x00000402)
memory[30] = 9216 (0x00002400)
memory[31] = 5 (0x00000005)
memory[32] = 258 (0x00000102)
memory[33] = 1034 (0x0000040A)
memory[34] = 14 (0x0000000E)
memory[35] = 10 (0x0000000A)
memory[36] = 55 (0x00000037)
memory[4092] = 4 (0x00000004)
memory[4094] = 34 (0x00000022)
memory[4095] = 55 (0x00000037)
```

20. fibonacci.log

```
fibonacci.log
1
2 --- 0 error(s), 0 warning(s) ---
3
```

21. fibonacci.lst

```
≡ fibonacci.lst
1  00000000 00100000 ldc 4096
2  00000001 0000000B a2sp
3  00000002 FFFFFFF0A adj -1
4  00000003 0000040D call main
5  00000004 0000010A adj 1
6  00000005 00002400 ldc result
7  00000006 00000004 ldnl 0
8  00000007 00000012 HALT
9  00000008 main:
10 00000008 FFFFFC0A adj -4
11 00000009 00000103 stl 1
12 0000000A 00002300 ldc N
13 0000000B 00000004 ldnl 0
14 0000000C FFFFFFF01 adc -1
15 0000000D 00000203 stl 2
16 0000000E 00000000 ldc 0
17 0000000F 00000303 stl 3
18 00000010 00000100 ldc 1
19 00000011 00000403 stl 4
20 00000012 loop:
21 00000012 00000202 ldl 2
22 00000013 0000090F brz done
23 00000014 00000402 ldl 4
24 00000015 00000302 ldl 3
25 00000016 00000006 add
26 00000017 00000403 stl 4
27 00000018 00000303 stl 3
28 00000019 00000202 ldl 2
29 0000001A FFFFFFF01 adc -1
30 0000001B 00000203 stl 2
31 0000001C FFFFF511 br loop
32 0000001D done:
33 0000001D 00000402 ldl 4
34 0000001E 00002400 ldc result
35 0000001F 00000005 stnl 0
36 00000020 00000102 ldl 1
37 00000021 0000040A adj 4
38 00000022 0000000E return
39 00000023 N:
40 00000023 0000000A data 10
41 00000024 result:
42 00000024 00000000 data 0
```

22. sum.asm and sum.o

```
PS C:\Users\saich\OneDrive\Desktop\AI> ./asm sum.asm
Object : sum.o (44 words)
Listing : sum.lst
Result : 0 error(s), 0 warning(s)
PS C:\Users\saich\OneDrive\Desktop\AI> ./emu sum.o
Loaded 44 instructions

HALT reached
Final Registers:
A = 43
B = 43
PC = 6
SP = 4096

--- Memory Dump (non-zero locations) ---
memory[0] = 1048576 (0x00100000)
memory[1] = 11 (0x0000000B)
memory[2] = -246 (0xFFFFF0A)
memory[3] = 525 (0x000020D)
memory[4] = 266 (0x000010A)
memory[5] = 18 (0x0000012)
memory[6] = -758 (0xFFFFD0A)
memory[7] = 259 (0x0000103)
memory[9] = 515 (0x0000203)
memory[11] = 771 (0x0000303)
memory[12] = 9472 (0x0002500)
memory[13] = 4 (0x0000004)
memory[14] = 3 (0x0000003)
memory[15] = 514 (0x0000202)
memory[16] = 2 (0x0000002)
memory[17] = 7 (0x0000007)
memory[18] = 272 (0x0000110)
memory[19] = 2833 (0x0000B11)
memory[20] = 9728 (0x0002600)
memory[21] = 514 (0x0000202)
memory[22] = 6 (0x0000006)
memory[23] = 4 (0x0000004)
memory[24] = 770 (0x0000302)
memory[25] = 6 (0x0000006)
memory[26] = 771 (0x0000303)
memory[27] = 514 (0x0000202)
memory[28] = 257 (0x0000101)
memory[29] = 515 (0x0000203)
memory[30] = -4847 (0xFFFFD11)
memory[31] = 770 (0x0000302)
memory[32] = 11008 (0x0002B00)
```

```
memory[33] = 5 (0x00000005)
memory[34] = 258 (0x00000102)
memory[35] = 778 (0x0000030A)
memory[36] = 14 (0x0000000E)
memory[37] = 5 (0x00000005)
memory[38] = 10 (0x0000000A)
memory[39] = 20 (0x00000014)
memory[40] = 30 (0x0000001E)
memory[41] = 40 (0x00000028)
memory[42] = 50 (0x00000032)
memory[43] = 150 (0x00000096)
memory[4092] = 5 (0x00000005)
memory[4093] = 4 (0x00000004)
memory[4094] = 5 (0x00000005)
memory[4095] = 150 (0x00000096)
```

23. sum.log

```
sum.log
1
2 --- 0 error(s), 0 warning(s) ---
3
```

24. sum.lst

```
≡ sum.lst
1  00000000 00100000 ldc 4096
2  00000001 0000000B a2sp
3  00000002 FFFFFFF0A adj -1
4  00000003 0000020D call main
5  00000004 0000010A adj 1
6  00000005 00000012 HALT
7  00000006 main:
8  00000006 FFFFFD0A adj -3
9  00000007 00000103 stl 1
10 00000008 00000000 ldc 0
11 00000009 00000203 stl 2
12 0000000A 00000000 ldc 0
13 0000000B 00000303 stl 3
14 0000000C loop: |
15 0000000C 00002500 ldc N
16 0000000D 00000004 ldnl 0
17 0000000E 00000003 stl 0
18 0000000F 00000202 ldl 2
19 00000010 00000002 ldl 0
20 00000011 00000007 sub
21 00000012 00000110 brlz loop_body
22 00000013 00000B11 br loop_done
23 00000014 loop_body:
24 00000014 00002600 ldc array
25 00000015 00000202 ldl 2
26 00000016 00000006 add
27 00000017 00000004 ldnl 0
28 00000018 00000302 ldl 3
29 00000019 00000006 add
30 0000001A 00000303 stl 3
31 0000001B 00000202 ldl 2
32 0000001C 00000101 adc 1
33 0000001D 00000203 stl 2
34 0000001E FFFFED11 br loop
35 0000001F loop_done:
36 0000001F 00000302 ldl 3
37 00000020 00002B00 ldc result
38 00000021 00000005 stnl 0
39 00000022 00000102 ldl 1
40 00000023 0000030A adj 3
41 00000024 0000000E return
42 00000025 N:
```

```
43 00000025 00000005 data 5
44 00000026 array:
45 00000026 0000000A data 10
46 00000027 00000014 data 20
47 00000028 0000001E data 30
48 00000029 00000028 data 40
49 0000002A 00000032 data 50
50 0000002B result:
51 0000002B 00000000 data 0
```

25.